

COURSE CODE: BUC 2101

DATE: 2/12/2015

SEMESTER: I

TIME: 9.00 - 12.00 noon

INSTRUCTIONS:

1. ATTEMPT ONLY FOUR (4) QUESTIONS

2. ALL QUESTIONS CARRY THE SAME MARKS

Question One:

- a) Explain the following phases of software development lifecycle
- i. Planning (4 Mark)
 - ii. Analysis (4 Mark)
 - iii. Design (4 mark)
 - iv. Implementation (4 Mark)
- b) Software requirements elicitation is one of the important activities in systems development lifecycle
- i. Explain the steps for designing and developing a good questionnaire (5 mark)
 - ii. Differentiate between "observation" and "site visit" fact-finding techniques (4 Marks)

Question Two:

- a) Briefly describe a complete computer-based information system in terms of its components (5 Ma)
- b) Justify with five reasons the importance of studying systems analysis and design (5 Ma)
- c) Explain any five qualities of a modern systems analyst (5 ms)
- d) Explain three fundamental objectives of systems maintenance (6 M)
- e) What role does the systems analyst play in system recovery? (4 M)

Question Three:

- a) Explain how social engineers manipulate fellow human beings (6 marks)
- b) What are the three areas of security problems in web-based systems? (6 marks)
- c) What roles do system designers, system builders and system owners play in information systems development project? (9 marks)
- d) Explain the term "Problem" in three different perspectives in information systems development (6 marks)

quality
positive

Question Four

- a) Using symbols of a structured approach to system analysis and design, illustrate the relationship between the following entities (2 marks each)

- i. Doctor: Patient
- ii. Chef: Food
- iii. Student: school bag
- iv. Library: book

- b) In security and cryptography, explain the difference between the following:

- i. Symmetric key algorithms and asymmetric key algorithms (4 marks)
- ii. Password and biometric based systems (3 marks)

- c) In designing a database, there are various processes that must be adopted to have a good database design. Discuss these design processes of a database. (10 marks)

Question Five

- a) Using illustrations explain the following in detail (12 marks)

- Reasons for Designing -
- organise info -
- determine entities

Question Five

a) Using illustrations explain the following in detail each)

a. Conceptual model

b. Data flow diagrams

c. Entity relationship diagram primary - foreign

d. Data dictionary - meta data

e. Data store - keep data

- Reasons for designing -

- organise info.

- determine entities (2 marks)

- specify primary key.

- draft suitable relationship.

- refine design.

- 2) System development methodologies can be categorized as Structural, Rapid and Agile. With illustrations explain three (3) system development methods that belong to the Rapid development (RAD) methodology. (15 marks)

- Iterative
- Prototyping

Question Six

- a) With an illustration, explain what is meant by the system boundary in computer input and output interface design. (5 marks)
- b) Explain the various human engineering factors that must be incorporated in the design of the computer input and output interfaces to assist the user of an information system. (10 marks)
- c) Networks must be designed before they are built in an organization. Discuss the various features that must be addressed to have a very successful network design. (10 marks)

- Accuracy
- Approaches
- Expense
- Built
- Storage

END